

DOSY ^1H NMR as an Accessible Technique for Characterizing Trends in Bottlebrush Polymer Molecular Weight

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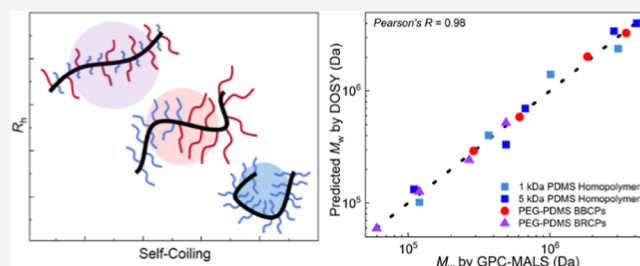


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ABSTRACT: Bottlebrush polymers (BBPs) make up a distinctive class of graft copolymers with densely packed side chains, whose unusual macromolecular architecture gives rise to unique physical properties with broad relevance across scientific disciplines. However, their branched architecture challenges traditional molecular weight characterization techniques. Gel permeation chromatography with multiangle light scattering (GPC-MALS) can accurately account for their branching, but this technique is not universally available. Despite its prevalence as a molecular weight determination methodology for linear polymer systems, diffusion-ordered spectroscopy (DOSY) remains underutilized for BBP characterization. Here, we show that DOSY offers an accessible alternative for determining trends in BBP molecular weight, achieving predictive relationships with high linearity ($R > 0.98$) across a breadth of compositions. Additionally, we show that DOSY can provide insights into unique conformational behavior of various BBP compositions through observed scaling relationships, further showcasing the technique's applicability.



INTRODUCTION

Bottlebrush polymers (BBPs) and bottlebrush block copolymers (BBCPs) make up an intriguing class of graft copolymers characterized by densely grafted side chains covalently attached to a central backbone. Their unique architecture, reminiscent of a plant of the same name, introduces a range of interesting properties and molecular parameters not found in conventional linear polymers. These properties include significant suppression of entanglements,^{1,2} rapid and well-ordered self-assembly over an impressive range of sizes and morphologies,^{3–10} and sterically induced conformational changes that have dramatic impacts for their applications as elastomers,¹¹ photonic materials and pigments,^{12–15} compatibilizers,¹⁶ and nanostructure templates.¹⁷ As a result, BBPs have gained significant attention in recent years, with interest likely to continue as new applications and chemistries emerge. However, the same molecular architecture that distinguishes BBPs from their linear counterparts also creates challenges for their characterization, specifically determining their molecular weights and solution conformations.

BBPs and BBCPs are commonly synthesized via grafting through polymerization of “macromonomers,” or low-molecular-weight polymers (<10 kDa) bearing a polymerizable functional group (*i.e.*, norbornene or vinyl groups) that form the chemical backbone once polymerized. Therefore, depending on the degree of polymerization of the macromonomer side chain (n_{sc} , Figure 1a), BBPs with low degrees of polymerization (DPs) of the backbone (denoted n_{bb} , Figure

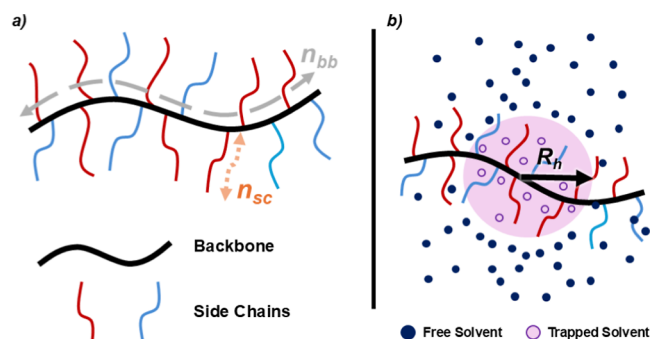


Figure 1. (a) Bottlebrush schematic highlighting important architectural parameters for BBPs in this study, including n_{bb} and n_{sc} . (b) Depiction of the hydrodynamic radius (R_h) of BBPs in solution portraying solvent flow around the polymer.

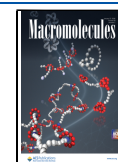
1a) can easily reach weight-average molecular weights (M_w) well over 100 kDa. Thus, ionization techniques such as matrix-assisted laser desorption/ionization time-of-flight (MALDI-ToF) mass spectroscopy are impractical, as the polymer is too

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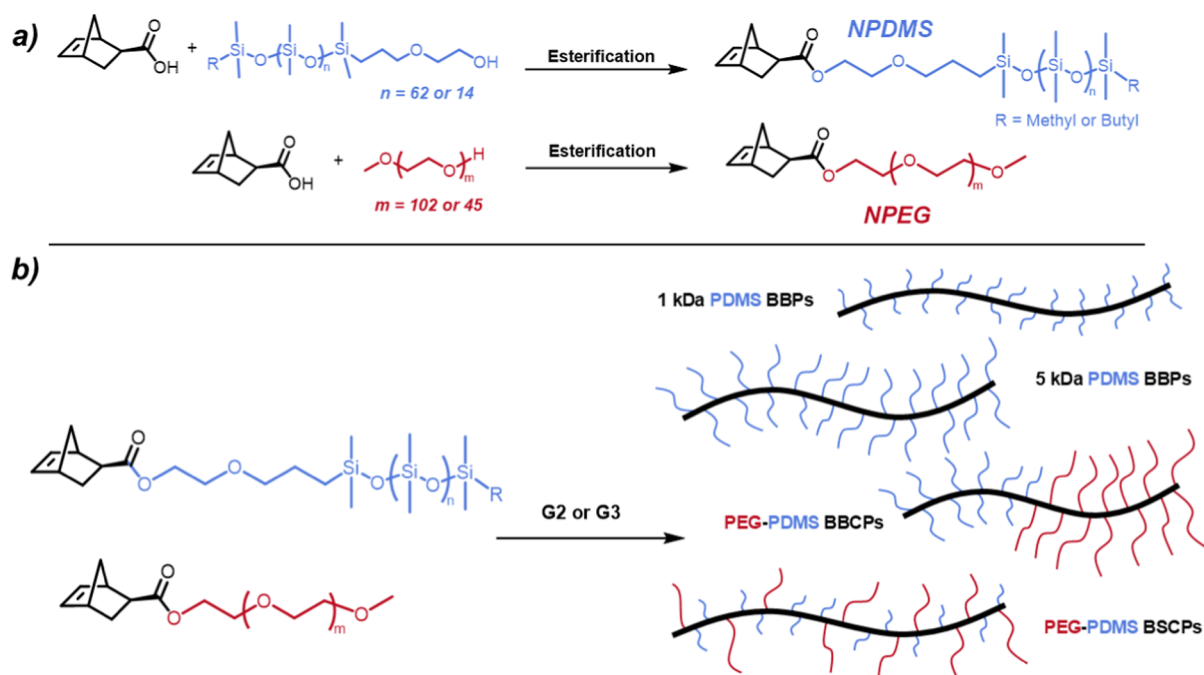
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Scheme 1. (a) The Coupling Reactions Used to Synthesize All NPDMS (Blue) and NPEG (Red) Macromonomers Used during This Study; Esterifications Were Performed Using Either the Mitsunobu Reaction or DCC Coupling; (b) Generalized Scheme for the Ring-Opening Metathesis Polymerization (ROMP) of the Synthesized Macromonomers



large to be desorbed from the substrate. Similarly, since the backbone and end-group typically make up a small percentage of the total weight of the polymer, end-group analysis via Nuclear Magnetic Resonance spectroscopy (NMR) is challenging, as these signals become dominated by the signals of side chains (see Figure S3). While light scattering techniques allow for accurate M_w determination, they require specialized equipment and numerous data points at various concentrations, increasing analysis time. Viscometric techniques, although successfully deriving relationships involving M_w for select bottlebrush systems,¹⁸ typically require large quantities of the polymer of interest and prove complex for phase-segregated BBCPs.¹⁹

Traditional gel permeation chromatography (GPC), the most common method for polymer M_w determination, requires additional considerations for BBPs as detailed by Walsh et al.²⁰ For BBPs of sufficient length, the dispersity measured in GPC is only characteristic of the backbone distribution. Thus, when side chains are grafted from or to the backbone, GPC does not provide information regarding grafting density or side-chain M_w distribution. Additionally, for BBCPs, monomodality is not sufficient to ensure the purity of the produced polymer, as factors such as refractive index differences, hydrodynamic scaling, and the separation resolution play important roles. Thus, researchers must often resort to GPC techniques equipped with multiangle light scattering (GPC-MALS),^{21–29} which combines light scattering principles with size-exclusion chromatography. While GPC-MALS accurately determines molecular weight distributions (MWDs) for branched architectures, its high cost can create a significant barrier to entry in BBP research.

Diffusion-ordered spectroscopy (DOSY) is a two-dimensional NMR technique commonly performed using standard NMR spectrometers, which are ubiquitous among chemistry institutions. Pointedly, DOSY can even be performed with high

accuracy on low-field, low-power benchtop spectrometers.³⁰ DOSY and GPC both rely upon determining a polymer's hydrodynamic radius (R_h) in solution, contrasting static light scattering techniques which rely upon the polymer radius of gyration (R_g). The R_h of a polymer provides a measure of size in terms of hydrodynamic drag and is often inferred from the Stokes–Einstein equation.³¹ This relationship equates both the polymer and the solvent trapped within the coil to an impermeable sphere that has the same center-of-mass diffusion³² (Figure 1b). Thus, as polymer M_w increases, its R_h increases and the polymer experiences more drag (*i.e.*, it diffuses more slowly). We can probe this change by measuring the polymer's effective diffusion coefficient (D) by DOSY. The relationship between M_w and R_h (or D^{-1}) has been explored in thorough detail for linear polymer systems^{32–39} allowing researchers to determine both molecular weight and polymerization kinetics⁴⁰ with high fidelity.

Despite the challenges in determining BBP molecular weight and the widespread accessibility of DOSY, there has not been a systematic report of its use for this purpose. A key advantage of DOSY is its solvent versatility—it can be performed in virtually any solvent in small amounts, provided there is sufficient gradient strength to suppress the signal of nondeuterated solvents. GPC generally requires large solvent volumes for column washing and chromatogram acquisition. Notably, DOSY has also demonstrated high accuracy in determining the M_w of linear polymers with low-field (80 MHz) benchtop NMR spectrometers.³³ Furthermore, because NMR spectroscopy is already used to characterize polymer structure, the same sample can also be used for DOSY to determine molecular weight distribution, saving material and time.

Here, we demonstrate its utility through the synthesis of a 17-member BBP and BBCP library varying in molecular weight, composition, and architecture using PDMS and PEG side-chain chemistries. DOSY experiments were performed to

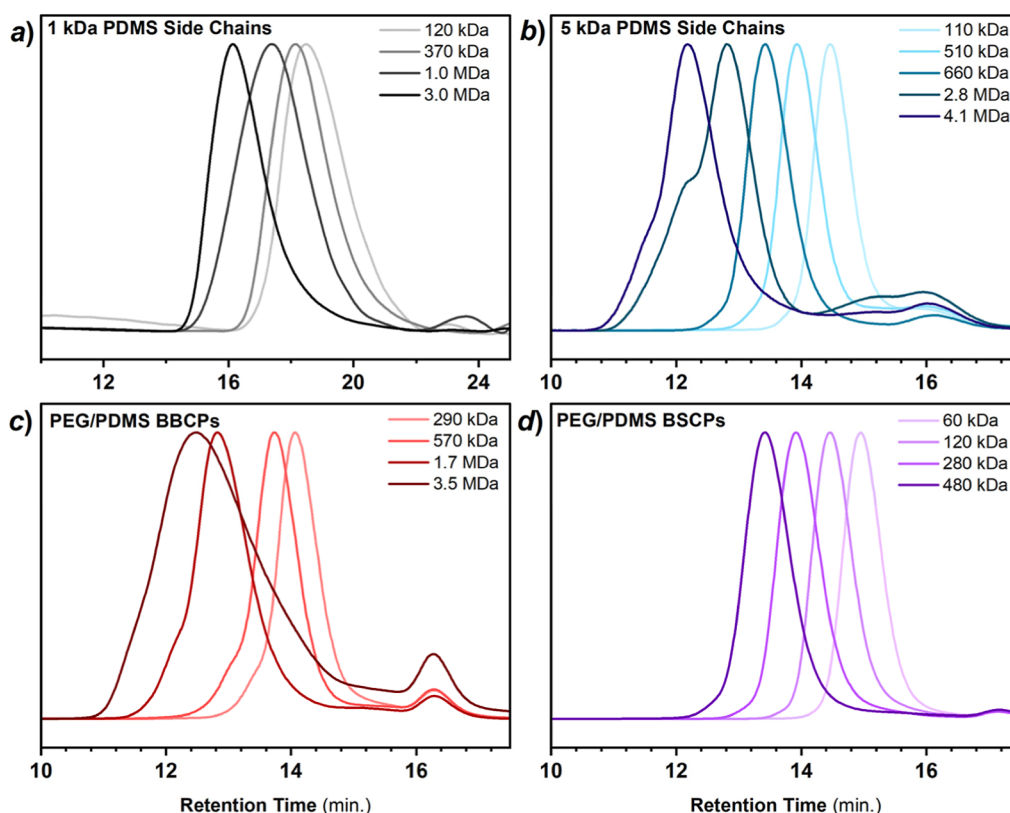


Figure 2. GPC chromatograms obtained for (a) PNB-*g*-PDMS₁₄ BBPs, (b) PNB-*g*-PDMS₆₂ BBPs, (c) (PNB-*g*-PDMS₆₂)-*block*-(PNB-*g*-PEG₁₀₂) BBPs, and (d) (PNB-*g*-PDMS₁₄)-*ran*-(PNB-*g*-PEG₄₅) BSCPs. All chromatograms were collected using THF as a solvent with polystyrene standards. Absolute molecular weights of the BBPs were determined by GPC-MALS in THF with 1 vol % triethylamine as the eluent. The smaller peaks remaining at higher elution times correspond to the macromonomer that was not functionalized with norbornene during the esterification step of their syntheses. Integration of these peaks shows <6% of these samples is reflected in these lower molecular weight peaks (Figure S4).

obtain diffusion coefficients, which were then correlated with molecular weights determined by GPC-MALS. Linearized models relating hydrodynamic radius (R_h) to known M_w values exhibited strong correlation ($R > 0.98$) across all architectures tested. These models also reveal scaling behavior that offers insights into the complex solution-state properties of the BBPs. This study highlights the potential of DOSY as an accessible and effective technique for BBP and BBCP molecular weight characterization.

BOTTLEBRUSH POLYMER ARCHITECTURES

The BBPs described here were synthesized via ring-opening metathesis polymerization (ROMP), allowing for excellent control over total molecular weight (M_w), side chain identity, side chain degree of polymerization (n_{sc}), backbone degree of polymerization (n_{bb}), and grafting density.^{3,17,41,42} The efficient and robust catalysts developed by Grubbs and Hoyveda⁴³ enable the rapid, controlled synthesis of a broad range of architectures. Accordingly, ROMP remains one of the most effective ways to synthesize BBPs and BBPs.¹⁷ To synthesize monomers polymerizable via ROMP, monohydroxy-terminated poly(dimethylsiloxane) (PDMS) and poly(ethylene glycol) (PEG) were coupled to norbornene carboxylic acid to yield ester-linked NPDMS and NPEG macromonomers, respectively (Scheme 1). These NPDMS and NPEG macromonomers were selected for their strong usage by bottlebrush researchers to synthesize bottlebrush elastomers^{6,11,44–49} and self-assembled nanostructures,^{17,50–55} among others.^{56–59} In total, four macromonomers were synthesized: 1 kDa NPDMS,

5 kDa NPDMS, 2 kDa NPEG, and 5 kDa NPEG. Detailed synthetic information about their coupling can be found in the Supporting Information. To ensure reproducibility and applicability, BBPs and BBPs were synthesized with different macromonomers, catalysts, and reaction conditions, yielding a library of 17 different samples spanning four different polymer compositions with M_w values ranging from 60 to 4000 kDa (Scheme 1).

To establish comparative benchmarks, PDMS homopolymer brushes with 1 kDa side chains were synthesized using the second-generation Grubbs Catalyst (G2), affording four polymers with M_w ranging from 120 to 3040 kDa (n_{bb} between 70 and 1840) (Figure 2). Higher dispersities ($\bar{D} = 1.5$) were generally observed in these samples due to the slower initiation of G2 compared to its third-generation counterpart (G3)⁶⁰ and due to catalyst deactivation over long reaction times.⁶¹ Another series of PDMS homopolymer brushes was synthesized using 5 kDa NPDMS macromonomers. These polymerizations were performed in an Ar-filled glovebox and utilized G3 to produce 5 samples with M_w ranging from 100 to 4000 kDa. These samples had lower dispersities due to the fast initiation of G3, although slight shoulders are apparent in the sample GPC chromatograms. We attribute this high-molecular-weight deviation to the presence of small amounts of macromonomers bearing two norbornene end groups,^{62,63} causing the formation of minute quantities of dimerized bottlebrushes. However, these impurities make up a minority fraction of the total sample, and thus we assume have little impact on the bottlebrush diffusive behavior.

In addition to the PDMS BBPs, we synthesized BBCPs and bottlebrush statistical copolymers (BSCPs). The BBCPs contained 5 kDa PDMS and 5 kDa PEG side chains, while the BSCPs used 1 kDa PDMS and 2 kDa PEG macromonomers. ROMP of these macromonomers using G3 produced BBCPs in the range of 290 kDa to 3.5 MDa with high conversion and generally low dispersity ($\bar{D} < 1.2$). It should be noted that small amounts of the nonfunctionalized side chain (e.g., hydroxy-terminated PEG or PDMS) were present in the final product corresponding to the peak seen at 16.5 min in the GPC chromatogram. However, this peak represents a small (<6%) amount of the final product (Figure S4), and we do not expect it to substantially influence the diffusive behavior of the BBCPs. Lastly, the BSCPs were synthesized with G3 using a Schlenk apparatus to afford a set of four polymers with M_w ranging from 50 to 480 kDa and low dispersities ($\bar{D} < 1.1$). High monomer conversion was achieved in all cases, and the desired M_w and dispersities aligned with the expected values. A summary of the synthesized polymers is given in Table 1.

Table 1. Summary Table of Samples Analyzed by GPC-MALS and DOSY^a

sample name	M_w (kDa)	$\bar{D}$	M_n (kDa)	D (10^{-10} m ² /s)	R_h (nm)
PNB ₇₀ -g-PDMS ₁₄	120	1.5	80	0.59	6.9
PNB ₂₂₅ -g-PDMS ₁₄	370	1.5	250	0.36	10.5
PNB ₆₁₀ -g-PDMS ₁₄	1010	1.5	670	0.23	16.2
PNB ₁₈₄₀ -g-PDMS ₁₄	3040	1.5	2020	0.19	20.9
PNB ₂₀ -g-PDMS ₆₂	110	1.1	100	0.58	6.5
PNB ₈₅ -g-PDMS ₆₂	510	1.1	470	0.41	9.2
PNB ₁₀₀ -g-PDMS ₆₂	660	1.1	590	0.31	12.6
PNB ₂₉₀ -g-PDMS ₆₂	2840	1.4	2060	0.17	22.8
PNB ₅₁₅ -g-PDMS ₆₂	4110	1.3	3280	0.16	23.4
(PNB ₂₅ -g-PDMS ₆₂)- <i>block</i> - (PNB ₂₅ -g-PEG ₁₀₂)	290	1.1	270	0.37	10.2
(PNB ₅₀ -g-PDMS ₆₂)- <i>block</i> - (PNB ₅₀ -g-PEG ₁₀₂)	570	1.1	530	0.28	13.4
(PNB ₁₄₀ -g-PDMS ₆₂)- <i>block</i> - (PNB ₁₄₀ -g-PEG ₁₀₂)	1770	1.1	1580	0.17	21.8
(PNB ₂₂₀ -g-PDMS ₆₂)- <i>block</i> - (PNB ₂₂₀ -g-PEG ₁₀₂)	3460	1.6	2230	0.14	26.1
(PNB ₁₇ -g-PDMS ₁₄)- <i>ran</i> - (PNB ₁₈ -g-PEG ₄₅)	60	1.1	60	0.71	5.3
(PNB ₃₆ -g-PDMS ₁₄)- <i>ran</i> - (PNB ₃₇ -g-PEG ₄₅)	120	1.1	120	0.50	7.5
(PNB ₇₆ -g-PDMS ₁₄)- <i>ran</i> - (PNB ₇₇ -g-PEG ₄₅)	280	1.1	270	0.37	10.3
(PNB ₁₃₈ -g-PDMS ₁₄)- <i>ran</i> - (PNB ₁₃₉ -g-PEG ₄₅)	480	1.1	450	0.26	14.5

^aThe M_w listed was determined by GPC-MALS, the $\bar{D}$ was determined via GPC, the M_n was determined by $M_w(\bar{D})^{-1}$, D was determined via DOSY, and R_h was determined using eq 1. Sample n_{bb} was determined by dividing the M_n by the molecular weight of the macromonomers used during each polymerization.

Following polymerization, all samples were subjected to MALS-GPC to evaluate M_w , followed by DOSY at 1 mg/mL in CDCl₃ by utilizing a Bruker AVANCE III 400 MHz NMR spectrometer. The ¹H resonance frequency was set to 400.13 MHz in a 5 mm broadband-observe Prodigy cryoprobe, 3 mm NMR tubes, a probe temperature of 293 K, and a stimulated echo pulse sequence with a bipolar gradient. The methyl ¹H signal of the PDMS side chains (0.09 ppm) was used to track diffusive behavior, as this signal is present on all polymer

samples and is notably sharp/distinct. All signal decay curves were fit using the Bruker Dynamics Center software, and an example analysis can be found in the Supporting Information (Figures S1 and S2). Specifically, a two-component fit was used to negate diffusive effects from residual unfunctionalized PDMS or PEG present following the esterification step. We note that care should be taken such that DOSY samples are at concentrations lower than the overlap concentration of synthesized bottlebrushes. As described by models and simulations by Borisov and co-workers,^{64,65} bottlebrush polymers in dilute solution exhibit a persistence length that scales linearly with the bottlebrush cross section, allowing the estimation of a sample's R_g (and thus overlap concentration) from variables such as side chain length and grafting density. Utilizing these models shows good agreement with the measured values of R_h . The reader is encouraged to refer to Figure S5 for details on the equations utilized.

SCALING RELATIONSHIPS AND SOLVENT QUALITY

Before investigating the bottlebrush polymer molecular weights by DOSY, it is important to understand the information that the derived scaling relationships provide. Namely, the scaling relationships seen in these types of experiments can inform us about the “quality” of solvent used in these systems as they pertain to enthalpic interactions. Theta solvents are described as solvents of sufficient quality such that two-body interactions (i.e., monomer–monomer-excluded volume interactions) appear to be screened, or nonexistent.^{31,66} This means that the polymer's characteristic length (R_g , R_h) scales with the molecular weight to the 0.50 power. Importantly, though, theta solvents are *not* enthalpically neutral to the polymer—there still lies some, albeit small, penalty for having the polymer chain surrounded by the solvent molecules (i.e., $\chi \neq 0$).^{31,66} When the polymer samples have a solvent to which it is truly enthalpically neutral (meaning that the monomers in the polymer chain have no preference of having an adjacent monomer or a solvent molecule next to it), the polymer is in an a thermal solvent. Thus, the polymer will attempt to form a truly random walk, but since monomer units cannot occupy the same space, the polymer chain is slightly swollen, meaning that it will scale with M_w to 0.588, better known as the Flory exponent.³¹ These distinctions are important to note as the bottlebrush architecture introduces additional entropic considerations not found in linear polymers, meaning that scaling relationships derived for linear polymers cannot necessarily be directly translated to their bottlebrush counterparts.

BOTTLEBRUSH MOLECULAR WEIGHT DETERMINATION BY DOSY

To investigate how the M_w values of these BBPs, BBCPs, and BSCPs relate to their hydrodynamic radii, we can start by applying DOSY to NPDMS macromonomers (Table 2). The diffusion coefficients of a series of NPDMS macromonomers were measured using DOSY and then converted to their respective R_h values as determined from the Stokes–Einstein equation. These hydrodynamic radii were then plotted against their M_w values as seen in GPC (Figure 3a). As expected, higher M_w macromonomers experience more drag, effectively increasing its R_h (or decreasing its D). This relationship is quantified by the Stokes–Einstein Equation (eqs 1–3),^{32,36} resulting in the following linearized and predictive equations:

Table 2. Characterization Data for Macromonomers in Figure 3

GPC		diffusion-ordered spectroscopy (DOSY)		
M_w (kDa)	$\bar{D}$	D (10^{-10} m ² /s)	R_h (nm)	M_w (kDa)
1.1	1.1	4.84	0.84	1.1
4.6	1.6	2.26	1.80	4.7
10.1	1.5	1.45	2.81	10.9
12.1	2.0	1.43	2.85	11.2

$$R_h = \frac{k_b T}{6\pi\eta D} = A \times (M_w)^\alpha \quad (1)$$

$$\ln(R_h) = \ln(A) + \alpha \times (\ln(M_w)) \quad (2)$$

$$\ln(M_w) = \frac{\ln(R_h) - \ln(A)}{\alpha} = \frac{1}{\alpha} \ln\left(\frac{R_h}{A}\right) \quad (3)$$

where k_b is the Boltzmann constant, T is the absolute temperature, η is the solvent viscosity (0.536 mPa s for CHCl₃ at 25 °C),⁶⁷ and D is the diffusion coefficient obtained from DOSY measurements. A is an empirical proportionality constant and $1/\alpha$ is the slope (m), which is analogous to the reciprocal of the fractal dimension (d_f) of the polymer.³² Note that this relationship between R_h and M_w is only valid for dilute solutions and when the molecule of interest is significantly larger than solvent molecules, two conditions are already present in polymer NMR spectroscopy.³² The linearized model derived in eq 3 for the PDMS macromonomers exhibited the expected trend with good linearity (Pearson's $R = 0.99$) (Figure 3b). From the derived fit, R_h scaled with M_w to the 0.52 power, suggesting that CDCl₃ is slightly better than a theta solvent for PDMS, aligning with the expected "good solvent" behavior.

Expanding this methodology to the BBPs and BBCPs showed not only similar success but also interesting solution-state behavior. Despite the vast architectural differences between the polymers, including differences in side chain M_w , chemical compositions, and synthetic conditions, all the produced fits showed excellent linearity (Pearson's $R > 0.98$, Figure 4). Deviations from the model may be a result of polymer dispersity, as the DOSY model used assumes an infinitely narrow molecular weight distribution.

Independently modeling each polymer series with their respective slopes and intercepts (Figure 4a–d) shows that

these models accurately measure molecular weight across multiple orders of magnitude and across different polymer architectures. Combining the data from all synthesized polymers and treating them as a single combined series reveals that this diffusive behavior remains linear for all tested bottlebrush architectures (Figure 5a). Treating each polymer series as their own, independent modeling reveals that this technique is accurate for a diverse set of architectures (Figure 5b). Note that at extremely high molecular weights, the R_h of a given polymer sample becomes insensitive to M_w (see eq 2), meaning that this methodology is not applicable. Fortunately, BBPs with M_w greater than 4 MDa, the largest used in this study, are generally uncommon in BBP research.

SELF-COILING PHENOMENA

The scaling relationships, as determined by the slopes of the different polymer systems, provide insights into complex solution-state bottlebrush physics. In the case of the NPDMS macromonomers, the polymer coils can freely diffuse throughout the solvent, allowing a slightly smaller-than-ideal polymer chain conformation as seen by the slope of 0.52. However, once these macromonomers are tethered to a central backbone, as in the case of the PDMS homopolymer BBPs, the slope changes dramatically. Once polymerized, the side chains experience a reduction in their conformational degrees of freedom as steric effects from adjacent side chains and covalent attachment to the backbone prevent the adoption of their isolated, single-chain conformation. Naturally, the high grafting density also means that PDMS side chains remain closely packed to one another once they are stitched together via the norbornene backbone. We hypothesize that this proximity and the reduction in degrees of freedom render the solvent, which is slightly enthalpically unfavorable to PDMS (as seen by scaling behavior lower than a true self-avoiding walk in Figure 3a), to screen the interactions between adjacent side chains (Figure 6). As a result, the polymer will prefer a more globular conformation, as seen by the reduction in the scaling relationship between R_h and M_w to 0.36 and 0.38 for the 1 and 5 kDa BBPs, respectively (Figure 4). For reference, the R_h of an impermeable sphere with a constant monomer density would scale as $M_w \propto 0.33$. We refer to this behavior as "self-coiling," as the scaling behavior suggests that the side chains aggregate and collapse in on themselves despite the high conformational penalty to do so. In other words, the slight

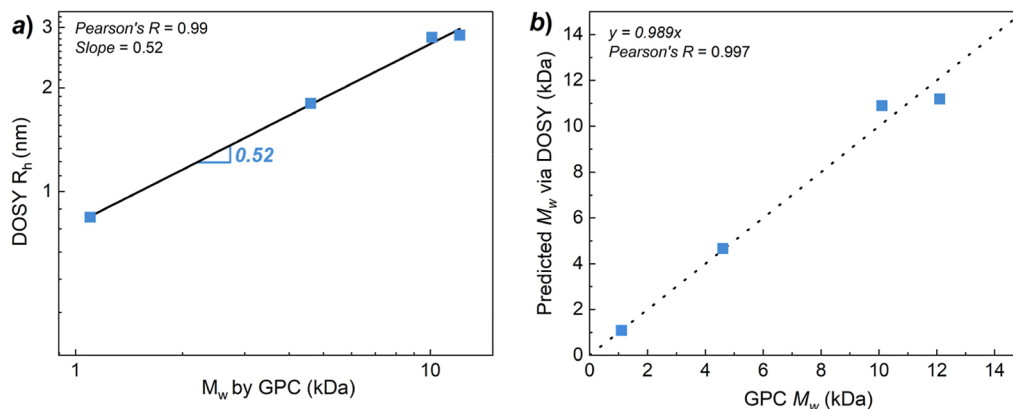


Figure 3. (a) Comparison of DOSY-derived R_h and GPC-derived M_w for a series of synthesized NPDMS macromonomers. (b) A comparison of the molecular weight of these macromonomers as determined by DOSY to those determined by GPC. A $y = x$ line illustrates the strong correlation.

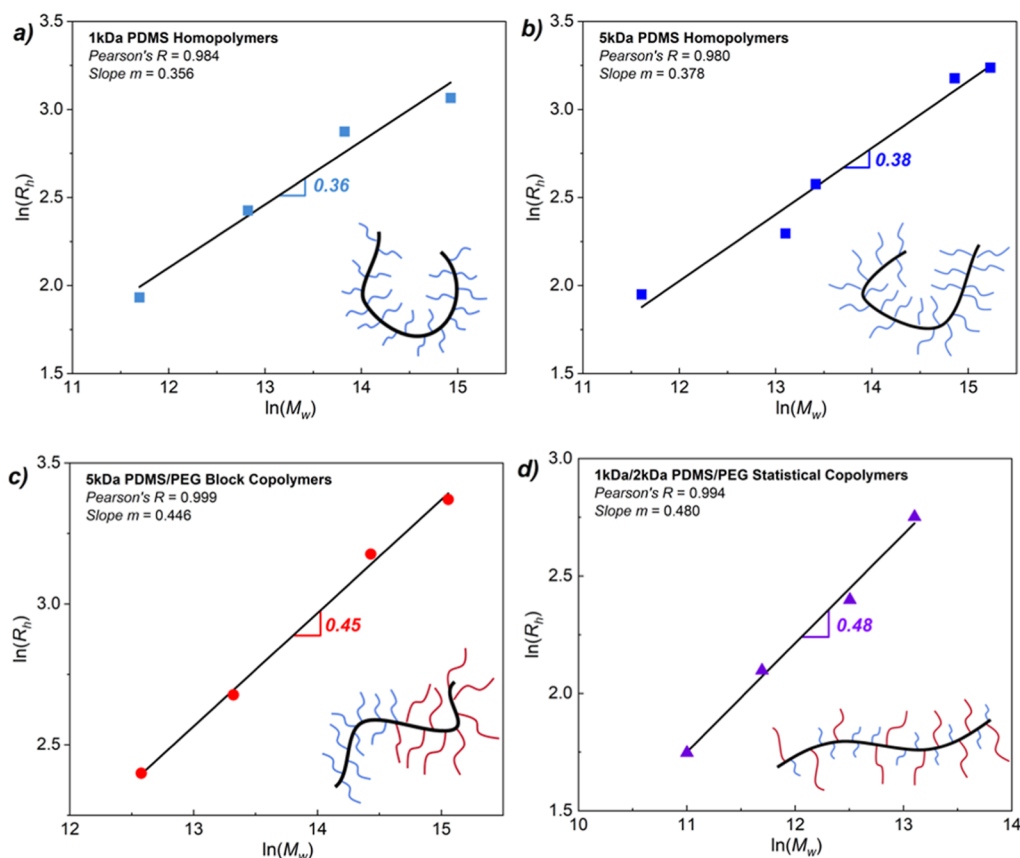


Figure 4. Linearized models comparing the R_h to the M_w of the synthesized (a) 1k PDMS homopolymer, (b) 5k PDMS homopolymer, (c) PDMS/PEG block copolymer, and (d) PDMS/PEG statistical copolymer bottlebrushes. Note the high correlations and increasing slopes as we alter the side chains of the bottlebrushes.

enthalpic mismatch between the side chains drives the side chains to one another, despite the larger entropic penalty required to do so.

These results differ but are not mutually exclusive from those published by Dutta et al., who characterized the solution-state behavior of poly[norbornene-*graft*-poly(lactic acid)] bottlebrush homopolymers in chlorobenzene,⁶⁸ using dynamic light scattering. Dutta et al. found that the R_h of their BBPs with varying side chain lengths scaled with M_w to 0.58 and 0.56 power for 5.2 and 2.3 kDa side chains, respectively. However, the increase in their observed scaling behavior can be attributed to the quality of the solvent, as the authors note that chlorobenzene is an athermal solvent for the poly(lactic) side chains. Thus, the tendency of their side chains to self-coil would be absent, leading to a more swollen coil conformation and an increased scaling exponent. Reduced scaling relationships similar to those seen in our work have also been observed by Leo et al. in poly[norbornene-*graft*-poly(styrene)] BBPs in dilute solution²² and in multiple other systems as highlighted by Sunday et al.⁶⁹ Generally, the observed scaling phenomena of bottlebrush polymers can be difficult to derive theoretically due to compounding and complex interactions. Foundational works by Birshtein predicted a scaling exponent of 3/5 for bottlebrush polymers in a theta solvent,⁷⁰ differing than the generally lower-observed values in the literature. This likely stems from assumptions of equal and negligible enthalpic interactions between the solvent and the polymer side chains and backbone, in addition to neglecting any enthalpic interactions between the side chains and backbone itself.

These important considerations are further complicated with the introduction of an additional side chain identity, as in the case with copolymers. Here, one must consider nonuniform swelling of the side chains, additional side chain–side chain enthalpic interactions (as in the case with our PEG–PDMS systems), and how all of these behaviors may change the effective “stiffness” or “blob size”.

Despite a similar scaling behavior for both the 1 and 5 kDa BBPs (Figure 4), the R_h measured via DOSY is notably different for a given n_{bb} (Figure 5c). We believe that these differences are attributed to steric constraints imposed by the longer, 5 kDa PDMS side chains. While both systems exhibit a tendency toward more collapsed conformations, the increase in side chain length may slightly hinder a complete collapse of the polymeric backbone into a sphere. Naturally, since the side chains appear to be dominating the conformation of the polymer, having larger side chains leads to a proportionally larger globule. Additionally, any 5 kDa PDMS side chains that exist outside of the globule make the polymer exhibit more drag, increasing R_h . These differences are most prominent when you compare PDMS BBPs of similar n_{bb} (Figure 5c). However, plotting the M_w (importantly not n_{bb}) vs R_h for all PDMS BBPs shows that the two homopolymers may exist on the same curve (Figure S5). Naturally, more side chain lengths should be tested, but this result suggests that the diffusive behavior of homopolymer bottlebrushes is dominated by the cumulative weight of the side chains, independent of the length of an individual side chain. Further, these relationships for homopolymer BBPs could be used to construct a “calibration

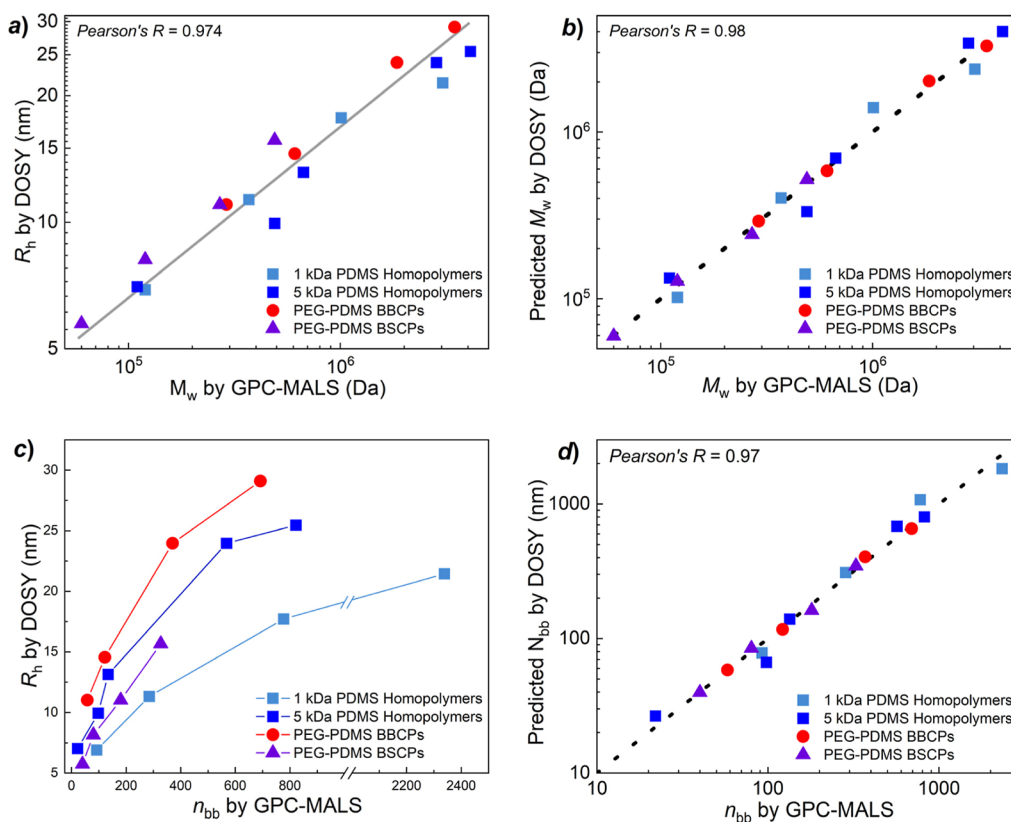


Figure 5. (a) Hydrodynamic radii of BBPs versus the M_w as recorded by GPC-MALS. (b) Estimated M_w with the DOSY model versus M_w recorded by GPC-MALS. A $y = x$ line is provided. (c) Hydrodynamic radii of BBPs versus n_{bb} . n_{bb} was calculated by dividing the total molecular weight of the polymer by the weight of its respective macromonomers. Lines are included to guide the eye. (d) Estimated n_{bb} from the DOSY model versus n_{bb} from GPC-MALS. A $y = x$ line is provided.

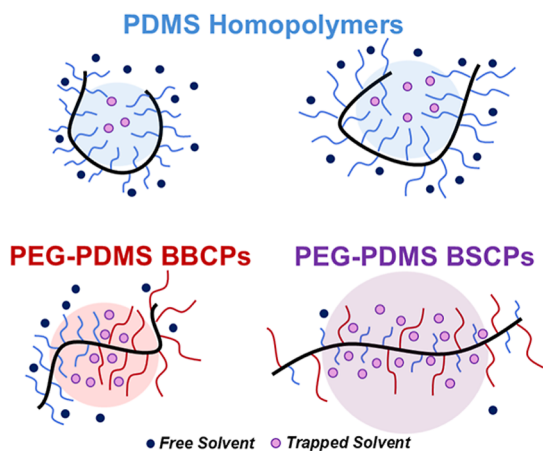


Figure 6. Illustration of the proposed “self-coiling” concept. As longer PDMS chains are introduced, the backbone is sterically hindered from adopting the collapsed hard sphere conformations possible with shorter side chains. The addition of additional side chains with dissimilar chemistries (in this case PEG) further confounds the ability of chains to self-aggregate in solution. This effect can be maximized by introducing a random copolymer architecture, where nearby PEG chains prevent PDMS chains from touching one another.

curve”, allowing researchers to estimate M_w solely from the measured diffusion coefficient irrespective of n_{sc} for a given side chain/solvent system. This apparent insensitivity to side chain length has also been seen in dynamic light scattering measurements by Dutta et al.⁶⁸

To investigate this phenomenon further, we introduced chemically dissimilar PEG side chains, which are known to unfavorably interact and phase separate with PDMS.^{45,47,50,51,55,71–73} These PEG side chains, due to their proximity to the PDMS side chains, were anticipated to prevent the collapse of BBCP into dense spheres. Specifically, placing these immiscible PDMS and PEG side chains next to one another should impose an enthalpic penalty too high for the phenomenon to occur. Thus, the backbone will favor coil expansion over collapse, increasing the scaling exponent observed as the chains swell, and these interactions begin to be screened.

Upon introduction of the PEG side chains in the BBPs, we observed a noticeable increase in the scaling behavior (or slope) from 0.38 to 0.45 for the 5 kDa PDMS BBPs and 5 kDa PEG/PDMS BBPs, respectively (Figure 4). This suggests that even though the PEG side chains only lie next to PDMS side chains at the interface between the two blocks, their inclusion is enough to alter the observed scaling behavior toward an expanded coil conformation. We similarly expected that increasing the number of PEG side chains near the PDMS side chains in a random copolymer architecture would cause an even higher degree of swelling. Accordingly, despite having the shortest side chain lengths, we observed the highest slope ($m = 0.48$) for the BSCPs (Figure 4). The absolute hydrodynamic size of these BSCPs was reduced compared to their block copolymer counterparts, as the PEG and PDMS side chains are substantially shorter (1 and 2 kDa versus 5 and 5 kDa, respectively). It is likely, then, that a BSCP with 5 kDa side

chains would have a larger R_h than the same side chains in a block configuration.

CONCLUSIONS

We have shown that DOSY and GPC-MALS can be successfully employed as molecular weight determination methods for a diverse set of bottlebrush polymers. Even for polymers with significant variations in side chain length, composition, and synthetic approaches, DOSY reliably predicted the molecular weight, producing linear calibration curves. We propose that differences in the scaling behavior between these samples arise from a self-coiling phenomenon, where the proximity of chemically similar side chains results in a more collapsed conformation, as side chain–side chain interactions may not be able to be screened by the solvent. This effect, however, can be mitigated by introducing chemically dissimilar macromonomers that disrupt self-coiling through unfavorable interactions, leading to coil expansion. Despite the broad range of structural differences among the polymer series, our findings demonstrate that DOSY is a reliable technique for determining both the M_w trends of and solution behavior for bottlebrush polymers. These results additionally emphasize DOSY's versatility and effectiveness in characterizing complex polymer architectures and helps democratize future research into this fascinating class of macromolecules. These results emphasize the potential of DOSY in the characterization of complex polymer architectures, encouraging future work to examine other difficult-to-characterize molecules such as zwitterions, complex coacervates, or simple sugars.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.macromol.5c01651>.

Details regarding synthesis (macromonomer and bottlebrushes), fitting of DOSY curve example, example of ^1H NMR spectra illustrating “washing out” of backbone proton signals in bottlebrush polymers, and GPC chromatogram integrating the remaining content of unconverted macromonomer following ROMP (PDF)

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Notes

The authors declare no competing financial interest.

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